UNIVERSIDAD DE PAMPLONA

Demostraciones de la Distribución Log-Normal

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Distribución log-normal

$$f_X(x) = \frac{1}{\sqrt{2\pi}\,\sigma x} e^{-\frac{1}{2}\left(\frac{\ln(x)-\mu}{\sigma}\right)^2}$$

$$\star\ E[X] = e^{\mu+\frac{\sigma^2}{2}}$$

$$\star\ \mathrm{Var}[X] = e^{2\mu+\sigma^2}(e^{\sigma^2}-1)$$

$$\star\ Z = \frac{\ln(x)-\mu}{\sigma}$$

Demostración

$$\int_0^\infty f_X(x)\,dx = 1$$

$$\star \int_0^\infty \frac{1}{\sqrt{2\pi}\,\sigma x} e^{-\frac{1}{2}\left(\frac{\ln(x)-\mu}{\sigma}\right)^2} dx$$

$$\star\ Z = \tfrac{\ln(x)-\mu}{\sigma}$$

$$\star\ dz = \tfrac{dx}{\sigma x} \Rightarrow dz = \sigma x\,dz$$

- $x = 0 \Rightarrow Z \to -\infty$
- $x = \infty \Rightarrow Z \to \infty$

$$\Rightarrow \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\cancel{\sigma x}} e^{-\frac{1}{2}z^2}\,\cancel{\sigma x}\,dz$$

$$\therefore \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}}\,dz = 1 \quad , \quad \text{Por distribución normal}$$

Demostración $E[X]$

$$\star\ E[X] = \int_0^\infty x \frac{1}{\sqrt{2\pi}\,\sigma x} e^{-\frac{1}{2}\left(\frac{\ln(x)-\mu}{\sigma}\right)^2} dx$$

$$\star\ Z = \frac{\ln(x)-\mu}{\sigma}$$

$$\star\ x = e^{\sigma z+\mu}$$

$$\star\ dz = \sigma e^{\sigma z+\mu}\, dz$$

$$\bullet\ x = 0 \Rightarrow Z \to -\infty$$

$$\bullet\ x = \infty \Rightarrow Z \to \infty$$

$$\Rightarrow \cdots = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\cancel{\sigma}} e^{-\frac{1}{2}z^2} e^{\sigma z+\mu} \cancel{\sigma} dz$$

$$\Rightarrow \cdots = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}z^2+\sigma z+\mu}\, dz$$

$$\star\ -\frac{1}{2}z^2 + 2\sigma z + \mu = -\frac{1}{2}(z^2 - 2\sigma z) + \mu$$

$$= -\frac{1}{2}((z-\sigma)^2 - \sigma^2) + \mu$$

$$= -\frac{1}{2}(z-\sigma)^2 + \mu + \frac{\sigma^2}{2}$$

$$\Rightarrow \cdots = e^{\mu+\frac{\sigma^2}{2}} \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z-\sigma)^2}\, dz\right]$$

$$\star\ w = z - \sigma$$

$$\star\ dw = dz$$

$$\bullet\ z = -\infty \Rightarrow w \to -\infty$$

$$\bullet\ z = \infty \Rightarrow w \to \infty$$

$$\Rightarrow \cdots = e^{\mu+\frac{\sigma^2}{2}} \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}w^2}\, dw\right]$$

$$\boxed{\therefore E[X] = e^{\mu+\frac{\sigma^2}{2}}}$$

Demostración $E[X^2]$

$$\star\ E[X^2] = \int_0^\infty x^2 \frac{1}{\sqrt{2\pi}\,\sigma x} e^{-\frac{1}{2}\left(\frac{\ln(x)-\mu}{\sigma}\right)^2}\,dx$$

$$\star\ Z = \frac{\ln(x)-\mu}{\sigma}$$

$$\star\ x = e^{\sigma z+\mu}$$

$$\star\ dx = \sigma e^{\sigma z+\mu}\,dz$$

$$\bullet\ x = 0 \Rightarrow Z \to -\infty$$

$$\bullet\ x = \infty \Rightarrow Z \to \infty$$

$$\Rightarrow \cdots = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\sigma z+2\mu} e^{-\frac{1}{2}z^2} e^{\sigma z+\mu}\,dz$$

$$\Rightarrow \cdots = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}z^2+2\sigma z+2\mu}\,dz$$

$$\star\ -\frac{1}{2}z^2 + 2\sigma z + 2\mu = -\frac{1}{2}[z^2 - 4\sigma z] + 2\mu$$

$$= -\frac{1}{2}[(z-2\sigma)^2 - 4\sigma^2] + 2\mu$$

$$= -\frac{1}{2}(z-2\sigma)^2 + 2\mu + 2\sigma^2$$

$$\Rightarrow \cdots = e^{2\mu+2\sigma^2}\left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z-2\sigma)^2}\,dz\right]$$

$$\star\ w = z - 2\sigma$$

$$\star\ dw = dz$$

$$\bullet\ z = -\infty \Rightarrow w \to -\infty$$

$$\bullet\ z = \infty \Rightarrow w \to \infty$$

$$\Rightarrow \cdots = e^{2\mu+2\sigma^2}\left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}w^2}\,dw\right]$$

$$\boxed{\therefore E[X^2] = e^{2\mu+2\sigma^2}}$$

Demostración $\mathbf{Var}[X]$

$$\star\ \mathrm{Var}[X] = E[X^2] - (E[X])^2$$

$$\Rightarrow \cdots = e^{2\mu+2\sigma^2} - \left(e^{\mu+\frac{\sigma^2}{2}}\right)^2$$

$$\Rightarrow \cdots = e^{2\mu+2\sigma^2} - e^{2\mu+\sigma^2}$$

$$\boxed{\therefore \mathrm{Var}[X] = e^{2\mu+\sigma^2}\left(e^{\sigma^2} - 1\right)}$$

